PRE-ALGEBRA (7TH) · LINEAR FUNCTIONS AND GRAPHING

Proportional Relationship Explorer

Explore & Reflect

Name _____ Date _____ Period _____

Open **Proportional Relationship Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. Trail mix costs $2 per pound. Is the total cost proportional to the weight? What if there is also a $3 bag fee?

2. Where does the graph of a proportional relationship cross the y-axis?

2 Explore use the tool

1. Set the price to $2.50 and drag the point to 1 pound. What is its height? Where else does that number appear?

2. Add a $4 bag fee. Record $\frac{y}{x}$ at 1, 2 and 4 pounds.

3. With the fee, does the cost still go up by the same amount each pound? Is that enough to make it proportional?

4. Find a price and a fee that make your store cost the same as Store B ($3 per pound) at 4 pounds.

3 Record use the tool and fill in the table

Compare stores

Price per pound	Bag fee	Equation	Cost of 6 lb	Proportional?
$1.50	$0			
$2.00	$3			
$4.00	$0			
$0.50	$6			

4 Explain what did you discover?

1. A relationship is proportional when $\frac{y}{x}$ is _____ for every pair.
2. Its graph is a straight line through _____ and its equation has the form _____.
3. The constant of proportionality k is the _____. It is the value of y when x = _____.
4. Explain why y = 2x + 3 has a straight-line graph but is not proportional.

5 Apply a new situation, no tool

1. A table shows (2, 7), (4, 14), (10, 35). Is it proportional? If so, find k and write the equation.
2. A table shows (1, 5), (2, 8), (3, 11). Is it proportional? Explain.
3. A proportional graph passes through (6, 9). Find k and the value of y when x = 10.